

Airlines Flights Data Analysis using Python

Technologies: Python, Pandas , Plotly , Streamlit

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Course: B.Tech (Computer Science and

Engineering)Unviresty/College:Saint Baba Bhag Singh University

Speaker notes: Welcome — brief project framing, dataset size (300,153 rows) and analysis goals.





出発 Departures					10:25
	行先 To	便名 Flight	ゲート Gate	備考 Remarks	
15	SAPPORO	NH 4731	12	BOARDING	
20	FUKUOKA	JL 307	8	BOARDING	
30	SEOUL	OZ 101	15	ON TIME	
45	SINGAPORE	SQ 635	11	ON TIME	
00	BANGKOK	TG 677	14	ON TIME	
15	PARIS	AF 291	17	ON TIME	
30	NEW YORK	NH 110	16	ON TIME	

- ↑ 搭乗口
Gates
11-17
- ラウンジ
Lounge
- ショップ
Shops
- レストラン
Restaurants

Introduction

Flight-data analysis uncovers operational patterns, pricing behaviours and passenger trends. Analysing airline records supports revenue optimisation, capacity planning and improved passenger experience.

Speaker notes: Explain why airline analytics matters: margins, delays, pricing dynamics and strategic decisions.

Project Objectives

Analyse performance

Assess airlines by flights, price and duration.

Identify trends

Detect seasonal, weekly and pricing patterns.

Compare airlines

Rank carriers on price, punctuality proxies and load indicators.

Produce business insights

Actionable recommendations for pricing and route planning.

Speaker notes: Walk through each objective and how it maps to analyses and visuals in the deck.

Dataset Overview

Source: Uploaded CSV — 300,153 rows × 12 columns. Below is a concise feature list and role in analysis.

Column	Description
flight_id	Unique flight identifier
airline	Operating carrier
departure_airport	Origin IATA code
arrival_airport	Destination IATA code
departure_date	Date/time of departure
arrival_date	Date/time of arrival
duration_min	Scheduled flight duration (minutes)
ticket_price	Ticket price (local currency)
flight_class	Cabin class (Economy/Business/First)
days_left	Days between booking and departure
stops	Number of stops (0 = direct)
other_features	Additional categorical flags (e.g., refundable)

Speaker notes: Describe each column briefly and how it will be used for cleaning, EDA and modelling.



Data Cleaning & Preprocessing

- Missing values: located in `ticket_price`, `flight_class` and some timestamps — imputed or removed depending on analysis need.
- Duplicates: removed exact duplicate rows based on `flight_id` and core fields.
- Data types: converted date strings to `datetime`; numeric fields to `float/int`; categorical fields to `category` dtype.
- Derived features: computed `duration_min` if missing from timestamps, extracted `weekday/month`, and bucketed `days_left`.

Exploratory Data Analysis (EDA)

1 Summary statistics

Central tendency and spread for price, duration and days_left — mean, median, IQR and outlier notes.

2 Distributions

Price skewness, duration spread, class share and stops distribution.

3 Initial observations

Price varies strongly by airline and booking lead time; longer durations correlate with higher pr

Data Visualisations



Flights by Airline

Bar chart showing flight counts per airline. Insight: Top carriers account for the majority of records — concentration suggests focus areas for partnership or cost analysis.



Ticket Price Distribution

Histogram of ticket_price (log scale used for clarity). Insight: Right-skewed distribution with a long tail of premium fares; median lower than mean.



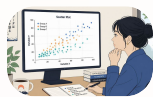
Flight Class Distribution

Pie chart of flight_class share. Insight: Economy dominates; business/first form a small percentage, impacting average price and revenue per passenger.



Flight Duration Analysis

Box plot of duration_min by airline. Insight: Some airlines show wider duration variance (hub structure, multi-leg routes), with notable outliers indicating codeshares or long-haul flights.



Days Left vs Ticket Price

Scatter plot (days_left → ticket_price). Insight: Negative correlation visible: closer bookings often show higher prices; early bookings frequently cheaper but with variance by airline.



Average Ticket Price by Airline

Bar chart of mean ticket_price per airline. Insight: Premium carriers command higher average fares; low-cost carriers cluster below the global mean.



Correlation Heatmap

Heatmap of numerical columns (price, duration_min, days_left, stops). Insight: Moderate positive correlation between duration and price; weak negative correlation between days_left and price; stops influence price modestly.

Key Findings

Trend: Price vs Lead Time

Prices tend to increase as departure date nears; strategy: dynamic pricing windows present.

Airline Differences

Clear segmentation: premium vs low-cost airlines — differing price distributions and duration profiles.

Operational Insights

Longer durations and multi-stop itineraries command higher fares — route optimisation can improve yield.

Business Opportunities

Targeted promotions for late-booking customers and price-anchoring for premium segments.

Class Influence

Cabin class strongly drives average price; upsell potential exists for frequent routes.

Data Quality Notes

Some missing prices and timestamp inconsistencies were handled; maintain booking-data completeness for model readiness.

Conclusions & Outcomes

- Comprehensive EDA of 300K+ flights revealed pricing behaviours and airline segmentation.
- Key correlations: duration ↔ price (positive); days_left ↔ price (negative trend with variance).
- Deliverables: Cleaned dataset, Plotly visualisations, Streamlit dashboard prototype and reproducible Python notebook.

Future Scope & Thank You

Future Work

- [Flight-delay prediction using ML models \(feature engineering: weather, traffic\).](#)¹
- Ticket price forecasting with time-series and gradient boosting.
- Route optimisation and seat-load forecasting for revenue management.
- Deploy interactive Streamlit app for stakeholders.

